

Front-End Performance Analysis

• Situation

This situation began when an internal client (the Director of Engineering Services at Northfield requested updated training for program managers (PMs) during the transition from iTLM to Wecenter (PLM). At the same time, a performance gap became visible: existing training modules were outdated and misaligned with new workflows. I also personally noticed discrepancies in role clarity and workflow execution, which compelled me to plan a structured front-end analysis before recommending solutions.

1. Background

- **Organization:** Northfield, a global leader in semiconductor equipment manufacturing.
- **Performance Situation:** Transition from iTLM to Wecenter has redefined PM roles and workflows. Current training content is outdated, leading to delays, role ambiguity, and compliance risks.
- **Workplace:** Distributed teams in the US and India, supported by JobHub LMS.
- **Work:** Managing engineering change requests (ECOs, CCNs, PCNs, CIMs) through SAP and Wecenter.
- **Workers:** Program managers with varying levels of system fluency and experience.
- **World:** Industry pressures for faster product iterations and ISO compliance.
- **My Role:** As an external instructional designer and performance consultant, I will plan and lead the FEA, ensuring it is solution-neutral and evidence-based.

📊 Front-End Analysis Plan

Impacts on Worker, Work, Workplace

- **Worker:** PMs lack clarity on new responsibilities, which leads to inconsistent execution and frustration.
- **Work:** Engineering change requests stall because approval gates and workflows are misunderstood, causing delays and rework.
- **Workplace:** The broader transformation risks compliance failures and reduced efficiency if training and performance support remain outdated.

Planning the Analysis: 5W + 1H

Element	Plan
Who	Clients: Director of Engineering Services, Global Training Lead Stakeholders: PMs, IT leads, L&D team, site managers SMEs: High-performing PMs already using Wecenter Me: Lead the planning, data collection, and synthesis
What	• Analyze gaps between expected (SHOULD) and actual (IS) performance • Examine role clarity, workflow execution, and compliance checkpoints • Determine if gaps stem from training, environment, or motivation
When	• Phase 1 (Entry): Stakeholder interviews and document review • Phase 2 (Data Collection): Observations and surveys • Phase 3 (Synthesis/Reporting): Consolidate findings and prepare recommendations
Where	• US and India sites with high onboarding volume • Virtual interviews and surveys

	• JobHub LMS content review and workflow observations in Wecenter (PLM)/SAP
Why	• To reduce delays, rework, and compliance risks • To ensure training aligns with actual job demands • To support global consistency during transformation
How	• Methods: Interviews, surveys, observations, document reviews, performance metrics • Tools: Gap analysis, workflow mapping, role clarity surveys • Data: ECO cycle times, escalation frequency, survey feedback, training content review

Determining Training vs. Other Issues

- Use Gilbert's Behavior Engineering Model (BEM) to distinguish environmental vs. individual factors.
- Apply Mager & Pipe Flowchart to confirm whether gaps are skill deficiencies or systemic issues.
- Assess business impact (delays, compliance risks) to determine if fixing the gap is worth the investment.

3. Sources and Rationale

I plan to use the following models and tools:

- **Harless's Front-End Analysis (FEA):** Guides stakeholder interviews with the "13 Smart Questions."
- **Gilbert's Behavior Engineering Model (BEM):** Diagnoses whether gaps stem from environment or capability.
- **Rummler-Brache Process Model:** Maps workflow across organizational, process, and performer levels.
- **Van Tiem et al.'s HPT Framework:** Provides a structured process for organizational, performance, and cause analysis.
- **Mager & Pipe Flowchart:** Validates whether the performance gap is due to a skill deficiency or something else.

These models complement each other. Harless ensures I ask the right questions, Gilbert helps identify root causes, Rummler-Brache highlights workflow misalignments, Van Tiem provides a comprehensive framework, and Mager & Pipe confirms whether training is the right solution. Together, they ensure the analysis is strategic, solution-neutral, and evidence-based.